



Web Development Concepts

ASP.NET · HTTP · Architecture · Performance

Comprehensive Study Notes covering core ASP.NET and web fundamentals with real-world scenarios and examples.

STUDY GUIDE — 2024

- ◆ Q1 — IActionResult vs ActionResult
- ◆ Q2 — HttpContext: Request & Response
- ◆ Q3 — HTTPS vs HTTP
- ◆ Q4 — Clean URL & URL Mapping
- ◆ Q5 — URL Segments & Fragments
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QUESTION 01

Why do we use `IActionResult` instead of `ActionResult` ?

Core Difference

Feature	<code>IActionResult</code>	<code>ActionResult</code>
Type	Interface	Abstract class
Flexibility	✓ Very flexible – any return type	✗ Tied to concrete hierarchy
Testability	✓ Easy to mock	✗ Harder to mock
Multiple return types	✓ Can return Json, View, File, Redirect...	✗ Limited to <code>ActionResult</code> subtypes
Generic version	<code>IActionResult</code>	<code>ActionResult<T></code>

Key Principle

Program to an interface, not an implementation. `IActionResult` is the contract; specific result classes (`ViewResult`, `JsonResult`, `FileResult`, `RedirectResult`, etc.) are the implementations.

Real-World Scenarios

Scenario 1 — E-Commerce Product Endpoint

A `GetProduct(int id)` action must return:

- `200 OK` with JSON product data if found
- `404 Not Found` if product doesn't exist
- `401 Unauthorized` if the user isn't logged in

```
// ✓ Using IActionResult – clean & flexible
public IActionResult GetProduct(int id)
{
    if (!User.IsAuthenticated) return Unauthorized();
    var product = _repo.FindById(id);
    if (product == null) return NotFound();
    return Ok(product); // JSON response
}
```

```
// ✗ Using ActionResult forces one return shape
public ActionResult GetProduct(int id)
```

```
{  
    // Can't easily return different HTTP status types  
}
```

Scenario 2 — Unit Testing

With `ActionResult`, you can mock and test different response scenarios independently:

```
// In test project  
var result = controller.GetProduct(99) as NotFoundResult;  
Assert.NotNull(result); // easy assertion  
  
// You can also check:  
var okResult = controller.GetProduct(1) as OkObjectResult;  
Assert.Equal(product.Id, ((Product)okResult.Value).Id);
```

Scenario 3 — File vs View Response

```
public IActionResult Download(string fileName)  
{  
    if (!System.IO.File.Exists(path))  
        return NotFound(); // returns 404  
    return File(fileBytes, // returns binary file  
        "application/octet-stream", fileName);  
}  
// One method - two completely different response types!
```

WHEN TO USE `ActionResult<T>` (C# 7.1+)

Use `ActionResult<T>` in API controllers for Swagger/OpenAPI documentation benefits — it gives you both flexibility AND type safety for the success case, while still allowing different HTTP responses.

```
public ActionResult<Product> GetProduct(int id) {  
    var p = _repo.Find(id);  
    if (p == null) return NotFound();  
    return p; // implicit 200 OK with Product JSON  
}
```

✗ THE PROBLEM WITH ACTIONRESULT ALONE

`ActionResult` is an abstract class. When you override it, you're locked into its class hierarchy. If a 3rd party library returns a custom `IActionResult` that doesn't inherit from `ActionResult`, you *cannot* use it as your return type — violating the Open/Closed Principle.

QUESTION 02

What does `HttpContext` Request & Response consist of?

`HttpContext` is the central object in ASP.NET Core — it carries everything about the current HTTP transaction: the incoming request and the outgoing response.

HttpRequest — What the Client Sends

Property	Type	Description & Example
<code>Method</code>	string	HTTP verb — <code>GET</code> , <code>POST</code> , <code>PUT</code> , <code>DELETE</code>
<code>Scheme</code>	string	<code>https</code> or <code>http</code>
<code>Host</code>	HostString	<code>api.mystore.com:443</code>
<code>Path</code>	PathString	<code>/products/electronics</code>
<code>QueryString</code>	QueryString	<code>?page=2&sort=price</code>
<code>Query</code>	IQueryCollection	Parsed: <code>Request.Query["page"]</code> → "2"
<code>Headers</code>	IHeaderDictionary	Authorization, Content-Type, Accept, User-Agent...
<code>Cookies</code>	IRequestCookieCollection	Session token, preferences
<code>Body</code>	Stream	Raw request body (JSON, form data)
<code>Form</code>	IFormCollection	Form fields & uploaded files
<code>RouteValues</code>	RouteValueDictionary	<code>{controller="Products", action="Get", id=5}</code>
<code>Protocol</code>	string	<code>HTTP/1.1</code> or <code>HTTP/2</code>
<code>IsHttps</code>	bool	<code>true</code> if HTTPS connection
<code>ContentType</code>	string?	<code>application/json</code>
<code>ContentLength</code>	long?	Size of request body in bytes

HttpResponse — What the Server Sends Back

Property	Type	Description & Example
<code>StatusCode</code>	int	200 OK, 201 Created, 400 Bad Request, 404, 500...
<code>Headers</code>	IHeaderDictionary	Content-Type, Cache-Control, Set-Cookie, CORS...
<code>ContentType</code>	string	<code>application/json; charset=utf-8</code>
<code>ContentLength</code>	long?	Size of response body
<code>Body</code>	Stream	The actual response data being written
<code>Cookies</code>	IResponseCookies	Set-Cookie: sessionId=abc123; HttpOnly
<code>HasStarted</code>	bool	If headers have been sent (can't modify after)

Other HttpContext Members

Member	Purpose
<code>User</code> (ClaimsPrincipal)	Authenticated user identity & claims
<code>Session</code>	Server-side session data
<code>Items</code>	Per-request key-value store (middleware sharing)
<code>Connection</code>	Remote IP, port, local IP
<code>WebSockets</code>	WebSocket upgrade & management
<code>RequestServices</code>	DI container scoped to this request
<code>TraceIdentifier</code>	Unique ID for request logging/tracing
<code>Abort()</code>	Cancel the current request

REAL USAGE EXAMPLE

```
public IActionResult Profile()
{
    // From Request
    var userId = HttpContext.Request.Query["id"];
    var token = HttpContext.Request.Headers["Authorization"];
    var ip = HttpContext.Connection.RemoteIpAddress;
    var user = HttpContext.User.Identity.Name;

    // Modify Response
}
```

```
HttpContext.Response.Headers.Add("X-Custom-Header", "value");  
HttpContext.Response.Cookies.Append("LastVisit", DateTime.UtcNow.ToString());  
  
return Ok(new { userId, user, ip });  
}
```

QUESTION 03

What's the difference between HTTPS and HTTP?

📌 One-Line Definition

HTTP = HyperText Transfer Protocol — sends data in plain text.
HTTPS = HTTP **Secure** — wraps HTTP inside TLS (Transport Layer Security) to encrypt, authenticate, and protect data integrity.

Aspect	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HTTP + TLS/SSL (Secure)
Default Port	80	443
Data Encryption	✗ Plain text	✓ Encrypted via TLS
Authentication	✗ No server identity proof	✓ SSL Certificate validates server
Data Integrity	✗ Can be tampered (MITM)	✓ Tampering detected via MAC
Speed	✓ Slightly faster (no crypto overhead)	✗ Minor overhead (negligible with HTTP/2)
SEO	✗ Lower Google ranking	✓ Preferred by Google
Browser Indicator	"Not Secure" 🔒	Padlock 🔒
Required for	Nothing critical	PWA, Service Workers, Geolocation, Camera, HTTP/2
Use Cases	Local dev only	Any production app

🔒 How TLS Handshake Works (HTTPS)



🖥️ Attack Scenarios HTTP Allows

👤 **Man-in-the-Middle (MITM) Attack on HTTP**

- User logs into a bank site over HTTP at a coffee shop WiFi.
- The attacker on the same WiFi intercepts the raw packets.
- They can read the username & password in **plain text**.
- With HTTPS, all they see is encrypted gibberish.

SSL Stripping Attack

Attacker downgrades HTTPS to HTTP by intercepting the first request. Mitigation: **HSTS** (HTTP Strict Transport Security) header tells browsers to always use HTTPS.

```
// ASP.NET Core - enforce HTTPS
app.UseHttpsRedirection();
app.UseHsts(); // sends Strict-Transport-Security header
```

HTTPS IN ASP.NET CORE

```
// Program.cs
builder.Services.AddHttpsRedirection(options =>
{
    options.RedirectStatusCode = StatusCodes.Status308PermanentRedirect;
    options.HttpsPort = 443;
});
app.UseHttpsRedirection();
app.UseHsts();
```

QUESTION 04

Clean URL & URL Mapping — LinkedIn Article Style

🌐 Why Your URLs Are Killing Your UX (And SEO)

Have you ever seen a URL like this?

`https://shop.com/index.php?cat_id=5&prod_id=234&ref=banner_2024`

Compare it to:

`https://shop.com/electronics/laptops/macbook-pro-16`

Which one would *you* click? Which one would you trust?

❌ Dirty URLs vs ✅ Clean URLs

Bad (Dirty) URL	Good (Clean) URL
<code>/page.php?id=42</code>	<code>/about-us</code>
<code>/products?cat=3&sub=12</code>	<code>/electronics/phones</code>
<code>/articles?year=2024&month=06&slug=web</code>	<code>/blog/2024/06/web-tips</code>
<code>/user.aspx?userId=99&action=edit</code>	<code>/users/99/edit</code>

📌 What is URL Mapping?

URL Mapping (Routing) is the process of associating a URL pattern to a specific controller/action. In ASP.NET Core, this is done via the routing middleware.

Types of Routing in ASP.NET Core

1 Convention-Based Routing

```
app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");
// /Products/Details/5 → ProductsController.Details(5)
```

2 Attribute Routing Recommended for APIs

```
[ApiController]
[Route("api/[controller]")]
public class ProductsController : ControllerBase
{
    [HttpGet("{id}")] // GET /api/products/5
    public IActionResult Get(int id) { ... }

    [HttpGet("featured")] // GET /api/products/featured
    public IActionResult GetFeatured() { ... }

    [HttpPost] // POST /api/products
    public IActionResult Create(ProductDto dto) { ... }
}
```

3 Minimal API Routing ASP.NET 6+

```
app.MapGet("/products/{id}", (int id) => Results.Ok(repo.Find(id)));
app.MapPost("/products", (Product p) => Results.Created($"products/{p.Id}", p));
```

Benefits of Clean URLs

-  **SEO** — keywords in URL path boost search ranking
-  **Shareability** — users can read & predict the URL
-  **Security** — hides implementation details (no `.php`, no internal IDs leaked)
-  **UX** — memorable, typeable, linkable
-  **Maintainability** — URLs stay stable even if backend changes

ROUTE CONSTRAINTS

```
[HttpGet("{id:int:min(1)}")] // only positive integers
[HttpGet("{slug:regex^[a-z-]+$}")] // only lowercase slugs
[HttpGet("{date:datetime}")] // must be valid date
```

QUESTION 05

URL Segments & Fragments — with Real Examples

Full URL Anatomy

```
https://www.techstore.com/products/electronics/laptops?brand=apple&sort=price#customer-reviews
```

Part	Value	Name
https	Scheme/Protocol	How to communicate
www.techstore.com	Host / Domain	Which server
/products/electronics/laptops	Path	Resource location
products , electronics , laptops	Segments ←	Path split by "/"
?brand=apple&sort=price	Query String	Filter/options
#customer-reviews	Fragment ←	Page anchor / section

URL Segments — Deep Dive

A **segment** is each part of the URL path separated by `/`. Each segment identifies a resource level in a hierarchy.

```
https://api.github.com/repos/microsoft/aspnetcore/issues/123
```

Segment	Meaning
repos	Resource type: repositories
microsoft	Organization/owner name
aspnetcore	Specific repository
issues	Sub-resource: issues of that repo

Segment	Meaning
123	Specific issue ID

```
// Reading segments in ASP.NET Core
[Route("repos/{owner}/{repo}/issues/{id}")]
public IActionResult GetIssue(string owner, string repo, int id) { ... }

// Or from HttpContext:
var segments = Request.Path.Value?.Split('/');
// ["", "repos", "microsoft", "aspnetcore", "issues", "123"]
```

📌 URL Fragment — Deep Dive

A **fragment** (hash `#`) identifies a specific *section within a page*. It is:

- ✓ **Never sent to the server** — it stays in the browser only
- ✓ Used for **in-page navigation** (anchor links)
- ✓ Used heavily in **Single Page Applications (SPAs)** for client-side routing
- ✓ Used for **deep linking** to a specific section

Real Examples:

URL	Fragment	Purpose
<code>https://docs.microsoft.com/en-us/aspnet/core/mvc/overview#the-model</code>	<code>#the-model</code>	Jump to "The Model" section in the docs
<code>https://github.com/user/repo#readme</code>	<code>#readme</code>	Scroll to README section
<code>https://myapp.com/#/dashboard</code>	<code>#/dashboard</code>	SPA hash routing (Angular/Vue)
<code>https://shop.com/product/5#reviews</code>	<code>#reviews</code>	Jump to reviews section

⚠️ IMPORTANT NOTES

- Fragments are **client-side only** — the server never sees them
- SPAs use `HashRouter` (React) or hash-based routing to work without server config
- For SEO-friendly SPAs, use `BrowserRouter` (HTML5 History API) instead — this uses real URL segments, not fragments

QUESTION 06

Builder Pattern & Dependency Injection — Real Life Examples

The Builder Pattern

The Builder Pattern constructs a complex object step-by-step. Instead of one massive constructor, you chain methods to configure the object, then call `Build()`.

Real-Life Analogy — Coffee Shop Order

You don't say "give me a large iced oat milk caramel latte with extra shot and no foam" all at once in a constructor call. You specify each option step by step:

```
var coffee = new CoffeeBuilder()
    .SetSize("Large")
    .SetType("Latte")
    .SetMilk("Oat")
    .AddFlavor("Caramel")
    .AddShot()
    .NoFoam()
    .Build(); // ← creates the final Coffee object
```

Builder in ASP.NET Core — `WebApplicationBuilder`

```
// Program.cs
var builder = WebApplication.CreateBuilder(args); // Get the builder

// Configure services step-by-step
builder.Services.AddControllers();
builder.Services.AddDbContext<AppDbContext>(opt =>
    opt.UseSqlServer(builder.Configuration.GetConnectionString("Default")));
builder.Services.AddAuthentication(JwtBearerDefaults.AuthenticationScheme)
    .AddJwtBearer(opt => { /* configure */ });
builder.Services.AddScoped<IProductService, ProductService>();

var app = builder.Build(); // ← Build the final WebApplication

// Configure the pipeline
app.UseAuthentication();
app.UseAuthorization();
```

```
app.MapControllers();
app.Run();
```

Dependency Injection (DI)

DI is a design pattern where a class *receives* its dependencies from the outside rather than creating them itself. The DI Container manages the creation and lifetime of those dependencies.

Real-Life Analogy — Electricity at Home

Your TV doesn't generate its own electricity. It has a plug (the interface), and the power company (the DI container) provides electricity through the socket. You can switch from company A to B without changing the TV.

-  **Interface** = the socket standard
-  **TV (Consumer)** = the class that needs the dependency
-  **Power Company (Implementation)** = the concrete service
-  **DI Container** = the electrical grid that routes power

Service Lifetimes

Lifetime	Method	Description	Example
Singleton	<code>AddSingleton</code>	One instance for the whole app lifetime	Configuration, Cache
Scoped	<code>AddScoped</code>	One instance per HTTP request	DbContext, Unit of Work
Transient	<code>AddTransient</code>	New instance every time it's requested	Email sender, Validators

```
// Register (in Program.cs)
builder.Services.AddScoped<IOrderRepository, SqlOrderRepository>();
builder.Services.AddSingleton<ICacheService, RedisCacheService>();
builder.Services.AddTransient<IEmailSender, SmtplibEmailSender>();

// Consume (in Controller) – NO new keyword needed!
public class OrderController : ControllerBase
{
    private readonly IOrderRepository _repo;
    private readonly IEmailSender _email;

    // DI Container injects these automatically
    public OrderController(IOrderRepository repo, IEmailSender email)
    {
```

```
        _repo = repo;
        _email = email;
    }

    [HttpPost]
    public async Task<IActionResult> PlaceOrder(OrderDto dto)
    {
        var order = await _repo.CreateAsync(dto);
        await _email.SendAsync(dto.Email, "Order Confirmed", $"Order #{order.Id}");
        return Ok(order);
    }
}
```

💡 WHY DI? — BENEFITS

- 🧪 **Testability**: Swap real SQL repo with mock in unit tests
- 🔄 **Loose Coupling**: Controller doesn't know (or care) which DB you use
- 🗝️ **Single Responsibility**: Each class does one thing
- ♻️ **Reusability**: Same service injected into multiple controllers

QUESTION 07

Web Pages (Razor Pages) vs MVC — with Business Cases

📌 High-Level Overview

Aspect	Razor Pages	MVC
Paradigm	Page-centric (one page = one file pair)	Pattern-centric (Model–View–Controller)
Structure	<code>.cshtml</code> + <code>.cshtml.cs</code> (PageModel)	Controller + Action Method + View
Routing	File-based (<code>/Pages/Products/Index.cshtml</code> → <code>/Products</code>)	Attribute or Convention routing
Best For	CRUD, forms, simple pages	Complex apps, APIs, large teams
Separation of Concerns	Moderate (UI & logic in one PageModel)	Strong (Controller / View / Model clearly separated)
Learning Curve	Lower	Higher
Code Reuse	Partial views, Tag Helpers	Shared Controllers, Views, Areas
API Support	✗ Not designed for REST APIs	✓ Native via ControllerBase

📁 Business Case 1 — Internal HR Leave Management Portal

✅ Best Choice: Razor Pages

Context: A mid-size company needs an internal portal where employees submit leave requests and managers approve/reject them.

Why Razor Pages wins:

- Each page has a clear, self-contained purpose: `ApplyLeave.cshtml` , `MyLeaves.cshtml` , `Approve.cshtml`
- The file-based routing matches exactly how users think about the app
- The `PageModel` keeps UI and page-specific logic together — perfect for a small team
- Simple CRUD forms don't need MVC's full separation overhead
- Faster development with less boilerplate

```
// Pages/Leave/Apply.cshtml.cs
public class ApplyModel : PageModel
{
    [BindProperty] public LeaveRequest Leave { get; set; }

    public async Task<IActionResult> OnPostAsync()
    {
        if (!ModelState.IsValid) return Page();
        await _leaveService.SubmitAsync(Leave, User);
        return RedirectToPage("/Leave/Confirmation");
    }
}
```

Business Case 2 — Multi-Platform E-Commerce Platform

✓ Best Choice: MVC + Web API

Context: A large retailer needs a web storefront (browser), a mobile app (iOS/Android), a third-party seller API, and an admin dashboard — all sharing the same business logic.

Why MVC wins:

- One **ProductsController** serves HTML views (web), JSON (mobile app), and XML (B2B sellers) — just different content negotiation
- Strong separation: business logic lives in services, not in page-specific code
- Areas (**/Admin** , **/Seller** , **/Customer**) for organized large-scale routing
- Easy to version the API (**/api/v1/products** , **/api/v2/products**)
- Multiple developers can work on Controller / View / Model independently

```
// MVC Controller serving multiple clients
[ApiController]
[Route("api/v1/[controller]")]
public class ProductsController : ControllerBase
{
    // Mobile app → JSON
    // B2B client → XML (via Accept header)
    // Web → Redirect to View
    [HttpGet("{id}")]
    public async Task<IActionResult> Get(int id)
    {
        var product = await _service.GetByIdAsync(id);
        return Ok(product); // content negotiation handles format
    }
}
```

```
}  
}
```

Summary Decision Matrix

Scenario	Recommended
Simple website with mostly forms	Razor Pages
REST API + Multiple clients	MVC / Web API
Admin CRUD panels	Razor Pages
Large team, many features	MVC
Personal blog / portfolio	Razor Pages
SaaS product with mobile app	MVC + Web API

QUESTION 08

Content-Type in Response — What, Where, and Why

📌 What Is Content-Type?

Content-Type is an HTTP header in the response (and sometimes request) that tells the receiver **what format the body data is in** and optionally its character encoding.

Format: `type/subtype; parameter=value`

```
Content-Type: application/json; charset=utf-8
Content-Type: text/html; charset=utf-8
Content-Type: multipart/form-data; boundary=---WebKitFormBoundary
Content-Type: image/png
Content-Type: application/pdf
```

📊 Common Content-Type Values

Content-Type	Use Case
<code>application/json</code>	REST API responses (JSON data)
<code>text/html</code>	Web page responses (HTML markup)
<code>text/plain</code>	Plain text files or logs
<code>application/xml</code>	XML data (legacy APIs, SOAP)
<code>application/pdf</code>	PDF file downloads
<code>application/octet-stream</code>	Binary file download (generic)
<code>image/jpeg</code> , <code>image/png</code>	Image files
<code>multipart/form-data</code>	File uploads from HTML forms
<code>application/x-www-form-urlencoded</code>	HTML form POST data (no files)
<code>text/css</code>	CSS stylesheets
<code>application/javascript</code>	JS files

🏠 Real Usage Scenarios

Scenario 1 — API returning JSON

```
// ASP.NET Core auto-sets Content-Type for Ok(object)
public IActionResult GetUser(int id)
{
    var user = _repo.Find(id);
    return Ok(user); // Content-Type: application/json; charset=utf-8
}
```

Scenario 2 — Returning a PDF file

```
public IActionResult DownloadInvoice(int orderId)
{
    byte[] pdfBytes = _invoiceService.Generate(orderId);
    return File(pdfBytes,
        "application/pdf", // ← Content-Type
        $"invoice_{orderId}.pdf"); // ← suggests filename
}
// Browser sees Content-Type: application/pdf
// → opens PDF viewer or prompts download
```

Scenario 3 — Serving dynamic images

```
public IActionResult GetAvatar(string userId)
{
    byte[] imageBytes = _storage.GetAvatar(userId);
    return File(imageBytes, "image/jpeg"); // Content-Type: image/jpeg
    // Browser renders it as an image, not a download
}
```

Scenario 4 — Content Negotiation (API)

```
// Client sends: Accept: application/xml
// Server checks Accept header and responds with XML
// ASP.NET Core handles this automatically with:
```

```
builder.Services.AddControllers()  
    .AddXmlSerializerFormatters();
```

? Why Is Content-Type Critical?

- **Without it**, the browser guesses the format (MIME sniffing) — a security risk called *MIME type confusion*
- **JavaScript's `fetch()`** checks it to know whether to parse as JSON, text, or blob
- **Security headers** like `X-Content-Type-Options: nosniff` force browsers to respect it
- Determines how browsers render, display, or download the response

⚠ COMMON MISTAKE

Sending JSON data but setting `Content-Type: text/plain` — the browser and client libraries will **not** parse it as JSON automatically, causing silent bugs.

QUESTION 09

Minification, Web Bundle, Webpack & Lazy Loading — Performance Deep Dive

These are all techniques to reduce how much data travels over the network and when it travels — directly impacting page load time.

1. Minification

Removes all unnecessary characters from source code (whitespace, comments, long variable names) **without changing functionality**.

Before Minification

```
// Calculate total price
function calculateTotal(price, tax) {
  var total = price + (price * tax);
  return total;
}
```

After Minification

```
function calculateTotal(p,t){return p+(p*t);}
```

✓ Typical size reduction: **20–80%** for JS/CSS files.

Tools: **Terser** (JS), **cssnano** (CSS), **html-minifier** (HTML).

2. Web Bundling

Combines multiple files into **one file**, reducing the number of HTTP requests.

Without Bundling (HTTP/1.1 bottleneck)

Browser downloads: `jquery.js` + `bootstrap.js` + `site.js` + `utils.js` = **4 separate requests**, each with its own TCP overhead, queuing in HTTP/1.1.

✓ With Bundling

All 4 files merged into `app.bundle.js` = **1 request**. Saves round-trip latency × 3.

```
// ASP.NET Core bundling in bundleconfig.json
[{"
```

```

"outputFileName": "wwwroot/js/site.min.js",
"inputFiles": [
  "wwwroot/lib/jquery/jquery.js",
  "wwwroot/lib/bootstrap/bootstrap.js",
  "wwwroot/js/site.js"
],
"minify": { "enabled": true }
}]

```

⚙️ 3. Webpack

Webpack is a powerful **module bundler & build tool** for JavaScript applications. It goes far beyond simple file concatenation.

Feature	Description
Module Bundling	Resolves all <code>import/require</code> statements and bundles them
Code Splitting	Creates multiple chunks loaded on demand
Tree Shaking	Removes unused code from the final bundle
Loaders	Transforms files: TypeScript → JS, SCSS → CSS, Images → base64
Plugins	HtmlWebpackPlugin, MiniCssExtractPlugin, etc.
Dev Server	Hot Module Replacement (HMR) for instant refresh in dev

```

// webpack.config.js (simplified)
module.exports = {
  entry: './src/index.js',
  output: {
    filename: '[name].[contenthash].js', // cache busting
    path: path.resolve(__dirname, 'dist'),
  },
  optimization: {
    splitChunks: {
      chunks: 'all', // ← code splitting
    },
    minimize: true, // ← minification
  },
  module: {
    rules: [
      { test: /\.tsx?$/, use: 'ts-loader' },
      { test: /\.scss$/, use: ['style-loader', 'css-loader', 'sass-loader'] },
    ],
  },
};

```

```
},  
};
```

4. Lazy Loading

Instead of loading *everything* at startup, lazy loading loads resources **only when they're needed**.

Types of Lazy Loading:

Type	What's Deferred	Trigger
Route-based	Page/component JS bundle	User navigates to that route
Component-based	Heavy UI components	Component enters viewport or is needed
Image lazy loading	Images below the fold	User scrolls near them
Data lazy loading	API data for a section	User opens a tab or accordion

```
// React – Route-based lazy loading  
import React, { Suspense, lazy } from 'react';  
const Dashboard = lazy(() => import('./pages/Dashboard'));  
const Reports = lazy(() => import('./pages/Reports'));  
  
function App() {  
  return (  
    <Suspense fallback={<Spinner />}>  
      <Routes>  
        <Route path="/dashboard" element={<Dashboard />} />  
        <Route path="/reports" element={<Reports />} />  
      </Routes>  
    </Suspense>  
  );  
}
```

// Dashboard.js is only downloaded when user visits /dashboard!

```
<!-- HTML – Native image lazy loading -->  
  
<!-- Browser loads this image only when it's near the viewport -->
```

Combined Impact on Network Performance

Technique	Reduces	Typical Gain
Minification	File size	20–80% smaller files
Bundling	Number of HTTP requests	10–50 requests → 2–3 requests
Webpack (Tree Shaking)	Dead code in bundle	30–60% smaller app bundle
Lazy Loading	Initial load size	50–80% less JS on first load
Gzip/Brotli + Minification	Transfer size	Up to 90% smaller over wire

💡 PRO TIP: CONTENT HASH CACHE BUSTING

Webpack appends a hash to filenames (`app.a3f4b2.js`). When the file changes, the hash changes, forcing browsers to download the new version while still caching unchanged files indefinitely.

```
// webpack output config
output: { filename: '[name].[contenthash:8].js' }
// Produces: main.a3f4b2c1.js – cached forever by CDN/browser
// On change: main.9d7e3f12.js – auto-invalidated
```

🌐 WHY THIS MATTERS OVER THE NETWORK

The bottleneck is often not the server processing time, but the time to transfer files over the network (especially on mobile/3G). These techniques directly attack that bottleneck:

- **Minification + Gzip** → less bytes to transmit
- **Bundling** → fewer TCP round trips
- **Lazy Loading** → only transmit what's needed right now
- **Tree Shaking** → don't ship unused library code

Result: **Faster Time to Interactive (TTI)**, better Core Web Vitals, happier users.